BLOCK 4

Differential 1-Bit Feedback DAC

From an initial estimate to a verified transistor-level baseline

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Technology	SKY130A, 1.8 V MOS devices
Clock / load	4 MHz / 1 pF per DAC output
Final DAC	$W_n/W_p = 8/256\,\mu$m, $L = 0.15\,\mu$m
Interface	StrongARM + two B01 buffers + SR latch
Evidence	1,080 cases; 45 PVT points; two waveform reviews
Revision	1.1 / 2 November 2026

Result status. Function and timing meet the pre-layout campaign criteria that were defined. These results are not a reliability sign-off, post-layout verification, or proof of the dynamic range of the complete ADC.

Independent design portfolio. English.

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Revision note 1.1

Revision 1.1 (2 November 2026) only updates the reference to the system-level target, from about 99 dB to about 75 dB SNDR at OSR 80. No Block 4 data, device size, or conclusion has been changed.

Summary for the reader

Block 4 converts a one-bit decision into two reference voltages that swap places. The final circuit remains simple: four CMOS transmission gates. The harder work lay in choosing a size that is strong enough in the worst-case condition, ensuring that the decision source is not disturbed by its load, and defining precisely when the output may be considered settled.

The pre-simulation note estimated an ON resistance of about 463 Ω and a settling time of 3.20 ns for a 1 pF load. That figure came from illustrative MOS parameters. The SKY130 model then gave 4.10 kΩ at mid-signal for the initial size $W_n/W_p = 2/4\,\mu$m, and the actual settling reached 17.77 ns. The reference-selector architecture was therefore correct, but the hypothesis that the initial size was sufficient was rejected by the test results.

The $4/8\,\mu$m size improved the nominal condition but passed only 26 of 45 PVT points. The SF, 1.62 V, −40 °C case required 137.92 ns. Subsequent exploration led to the $8/256\,\mu$m size, which satisfies all 45 standalone points. The large PMOS width is a real cost of this solution: conduction improves, but the control load and parasitic effects increase.

Integration with the StrongARM comparator and the SR latch exposed a second problem. At a 1 mV input difference, the direct connection produced only six correct decisions out of 12 diagnostic cycles. The comparator without the latch and the circuit with ideal isolation were correct again. Two transistor-based B01 buffers were then inserted between the comparator and the latch. This change is supported by a loading diagnosis; its single physical mechanism is not claimed to have been fully isolated.

Final result from the Xschem export	Value
PVT campaign / passing cases	45/45 / 1080/1080
Cases with 1 pF DAC / without DAC	540 / 540
Maximum latch decision delay	5.498 ns
Q/QB reaching the 1% rail band, maximum	7.853 ns
DAC settling after control is stable, maximum	6.400 ns
Maximum steady-state error	1.254 µV
Unresolved decisions / unsettled edges	0 / 0

The review of the two worst waveforms reproduces the summary figures and checks hold during reset. The total time from the start of the clock edge until the DAC is stable reaches 12.711 ns on the SF waveform. That figure differs from the 6.400 ns settling measured after the gate is stable. This difference in time definitions must be carried into the interface design of the next block.

Design decision. Keep the $8/256\,\mu$m DAC, $L = 0.15\,\mu$m, and the two B01 buffers as the pre-layout function/timing baseline. Repeating optimization passes A–D is not supported by any new finding. The glitch budget, reference current, whole-transistor stress, noise/mismatch, and post-layout remain open.

1 Design objective and scope of the work

1.1 Role of Block 4 in the modulator

The parent project targets a second-order switched-capacitor delta-sigma modulator with a nominal 1.8 V supply, 4 MHz sampling, and OSR 80. The sampling relation gives

$$T_s = \frac{1}{f_s} = 250\,\text{ns}, \qquad B = \frac{f_s}{2\,\text{OSR}} = 25\,\text{kHz}. \tag{1}$$

This bandwidth is an architectural relation, not the measured bandwidth of a finished ADC. The dynamic-range target is a system target: revision 1.0 of this report cited about 99 dB, and on 2 November 2026 the project owner revised it to about 75 dB SNDR at OSR 80 (see the Block 5 report). This revision concerns only the system-level target; the results and limits of Block 4 are unchanged. One-bit feedback is part of the delta-sigma loop [1]; the results of a reference-selector block are not sufficient to state the performance of the complete loop.

Block 4 selects the reference, while the sampling network and charge transfer are handled by Block 5. The non-overlapping clock generator is Block 6. This division matters: the 1 pF load in this report is a test load per output, not evidence that the entire downstream switched-capacitor interface has been modeled.

1.2 Questions to be answered

This report assesses whether the reference selection is correct, whether settling meets the budget, whether the nominal pass survives PVT, and whether the circuit keeps working when driven by a transistor-level comparator and latch. The next question is whether the Xschem implementation reproduces the behavior previously tested through the automation netlist.

This document is a retrospective synthesis. The initial expectations are taken from the pre-simulation note [S00]; the conclusions are then traced to experimental results. Rationalizations made after the measurements are not presented as numerical predictions made beforehand.

Table 1: Test specifications and the limits of their interpretation.

Parameter	Value used	Meaning
Nominal supply	1.8 V	The PVT campaign uses 1.62, 1.8, 1.9 V.
Final nominal reference	1.2 / 0.6 V	Differs from the initial rail-to-rail 1.8 / 0 V example.
Reference under PVT	$2V_{DD}/3$, $V_{DD}/3$	Not an absolute 1.2 / 0.6 V reference at every supply.
Integration test load	1 pF per output	Does not yet include the switches, OTA, and Block 5 impedance.
DAC tolerance	0.1% of reference span	Per-output settling criterion.
Settling budget	10 ns after control	Not a 10 ns total clock-to-DAC time.
Latch decision budget	100 ns	Separate from the 1% rail requirement.

2 Theoretical basis and circuit implementation

2.1 Differential and common-mode transfer

Define the decision $D = Q \in \{0, 1\}$ and $\overline{D} = QB$. With $\Delta V_{\mathrm{REF}} = V_{\mathrm{REFP}} - V_{\mathrm{REFN}} > 0$, the ideal outputs are

$$V_P = DV_{\mathrm{REFP}} + (1 - D)V_{\mathrm{REFN}}, \tag{2}$$

$$V_N = DV_{\mathrm{REFN}} + (1 - D)V_{\mathrm{REFP}}, \tag{3}$$

$$V_d = V_P - V_N = (2D - 1)\Delta V_{\mathrm{REF}}, \tag{4}$$

$$V_{CM} = \frac{V_P + V_N}{2} = \frac{V_{\mathrm{REFP}} + V_{\mathrm{REFN}}}{2}. \tag{5}$$

At the nominal 1.2/0.6 V references, $V_d = \pm 0.6\,\mathrm{V}$, the peak-to-peak differential swing is 1.2 V, and the ideal common mode is 0.9 V. At the 1.62 V PVT point the references become 1.08/0.54 V, so $V_d = \pm 0.54\,\mathrm{V}$ and the common mode is 0.81 V.

Q	QB	V_P	V_N	V_d
0	1	V_{REFN}	V_{REFP}	$-\Delta V_{\mathrm{REF}}$
1	0	V_{REFP}	V_{REFN}	$+\Delta V_{\mathrm{REF}}$

The state $Q = QB$ is not part of the ideal operating table. During a transition, both controls are analog voltages with finite rise/fall times; for a moment both may partially turn on different paths. A logic table alone therefore does not explain reference current or glitch.

2.2 Four transmission gates and pin order

Each TG consists of one NMOS and one PMOS in parallel. This combination is used to compensate for the conduction weakness of each device type over different parts of the signal range [3]. The TG is on when the NMOS gate is high and the PMOS gate is low. The NMOS bulk is tied to GND; the PMOS bulk to VDD.

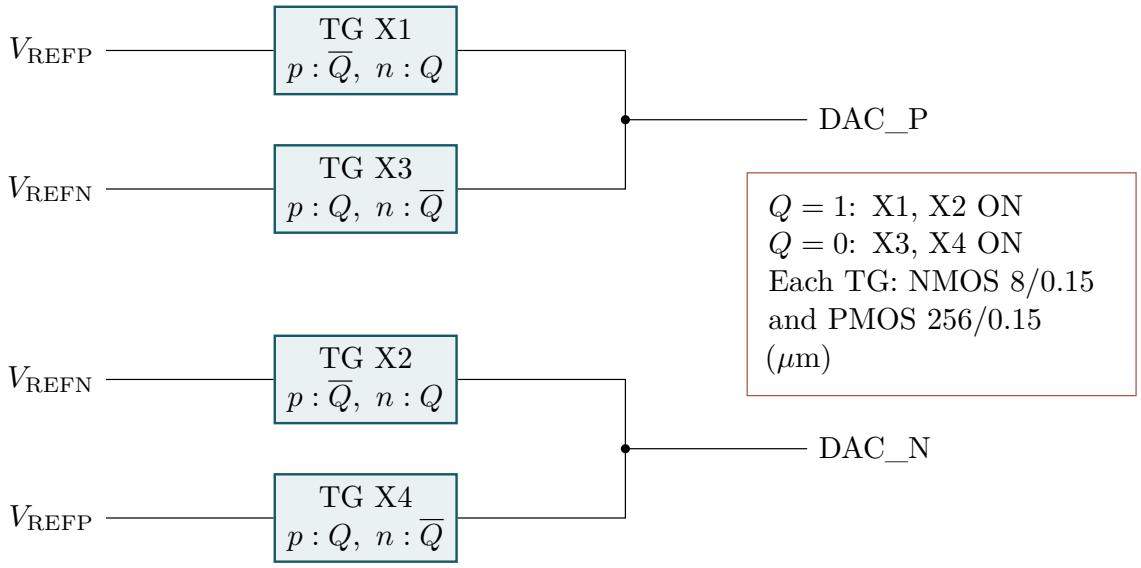


Figure 1: Functional connections of the final DAC. The numbers X1–X4 follow the tested export netlist; these labels are the reference that prevents numbering from being swapped between figures.

Table 2: Pin mapping taken from the final implementation [S22].

Instance	A	B	PMOS gate	NMOS gate
X1	VREFP	DAC_P	QB	Q
X2	VREFN	DAC_N	QB	Q
X3	VREFN	DAC_P	Q	QB
X4	VREFP	DAC_N	Q	QB

The TG subcircuit pin order is `A B CLK_P CLK_N VDD GND`. The DAC pin order is `VREFP VREFN Q QB VDD GND DAC_P DAC_N`. The DAC contains four NMOS and four PMOS devices, all with $L = 0.15\,\mu$m, `nf=1`, `mult=1`. The final sizes are $8\,\mu$m for each NMOS and $256\,\mu$m for each PMOS. This DAC cell is independent of the TG used by the CMFB; changing the DAC size does not automatically change the Block 2 cell.

2.3 Why the comparator needs a tested interface

The StrongARM produces a dynamic output that returns to its reset state every cycle. The SR latch stores the decision so that Q/QB remain available during that reset. The non-inverting B01 buffer uses two inverters per branch, so the polarity from the comparator to the latch input is unchanged.

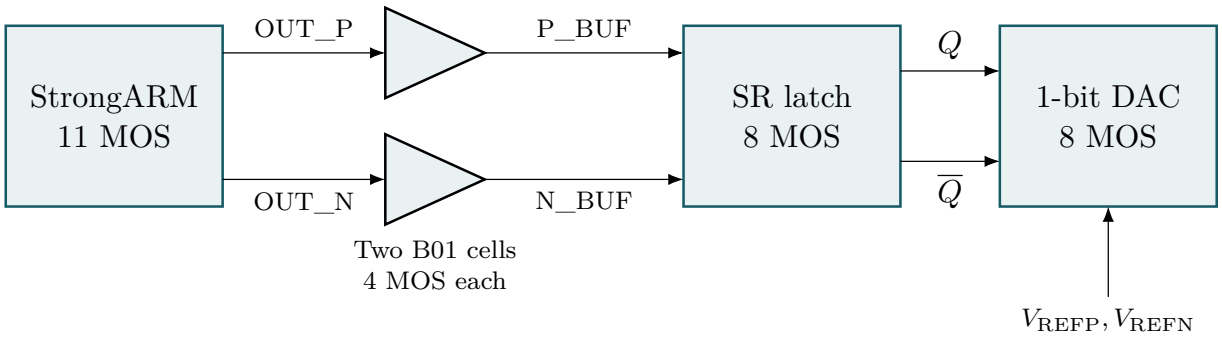


Figure 2: Final interface: 11 comparator MOS, 8 MOS for the two buffers, 8 latch MOS, and 8 DAC MOS; 35 MOS in total in the loaded configuration.

The buffer is placed *between the comparator and the latch.* It does not remove the DAC gate load that the latch Q/QB outputs must still drive. This separation explains why DAC characterization and interface testing remain two distinct pieces of work.

The latch core uses two cross-coupled NAND gates with active-low inputs SN and RN:

$$Q = \overline{SN \cdot QB}, \qquad QB = \overline{RN \cdot Q}. \tag{6}$$

With $SN = 0, RN = 1$, Q is set high; with $SN = 1, RN = 0$, Q is reset low. When the comparator resets and both inputs go high, the latch holds its previous state. Both inputs low is not an allowed storage state because it forces Q and QB high together. This is the physical reason retention must be checked, not assumed from the cell name.

As a local model of regeneration, the comparator output difference can be written $v_d(t) \approx v_{d0} \exp(t/\tau_{reg})$, so that

$$t_{reg} \approx \tau_{reg} \ln \frac{V_{decision}}{|v_{d0}|}, \qquad \tau_{reg} \sim \frac{C_{node}}{g_{m,reg}}. \tag{7}$$

This model predicts that a smaller initial signal or a larger node load can slow the decision. No g_m or C_{node} was extracted here to predict a numerical delay. From the netlist sizes, the NAND-gate proxy seen directly by each comparator output is $(2+2)0.15 = 0.60\,\mu\text{m}^2$, whereas the B01 input is only $(0.5+1)0.15 = 0.225\,\mu\text{m}^2$. The reduction of the proxy to 37.5% supports the choice of a small buffer; it is not a measurement of the actual capacitance.

3 Initial calculation: expectations and assumptions

3.1 ON-resistance model

To obtain an estimate before using the PDK model, the linear channel conductance is approximated by

$$g_n \approx \mu_n C_{ox} \frac{W_n}{L} \max(V_{\text{DD}} - V_{sig} - V_{Tn}, 0), \tag{8}$$

$$g_p \approx \mu_p C_{ox} \frac{W_p}{L} \max(V_{sig} - |V_{Tp}|, 0), \tag{9}$$

$$R_{\text{ON}} \approx (g_n + g_p)^{-1}. \tag{10}$$

This equation simplifies the effects of body bias, short channels, and voltage change along the transition. The initial note used $\mu_n C_{ox} = 200\,\mu\text{A/V}^2$, $\mu_p C_{ox} = 80\,\mu\text{A/V}^2$, and $V_{Tn} = |V_{Tp}| = 0.45\,\text{V}$. These values are *illustrative assumptions*, not SKY130 extraction results.

With $W_n/W_p = 2/4\,\mu\text{m}$, $L = 0.15\,\mu\text{m}$, $V_{\text{DD}} = 1.8\,\text{V}$, and $V_{sig} = 0.9\,\text{V}$:

$$g_n = (200 \times 10^{-6})(2/0.15)(0.45) = 1.20\,\text{mS}, \tag{11}$$

$$g_p = (80 \times 10^{-6})(4/0.15)(0.45) = 0.96\,\text{mS}, \tag{12}$$

$$R_{\text{ON}} \approx 1/(2.16\,\text{mS}) = 462.96\,\Omega. \tag{13}$$

The initial hypothesis was clear: if this resistance represents the path actually used, the inherited 2/4 size should be sufficient for the chosen settling budget. It is this hypothesis that was tested, not one assumed to be true because the equation looked reasonable.

3.2 RC settling derivation and resistance budget

For one branch with an ideal reference, constant R_{ON}, and capacitor C_L,

$$C_L \frac{dV_o}{dt} = \frac{V_{target} - V_o}{R_{\text{ON}}}. \tag{14}$$

The solution gives the normalized residual

$$e_n(t) = \exp[-t/(R_{\text{ON}} C_L)], \qquad t_\epsilon = R_{\text{ON}} C_L \ln(1/\epsilon). \tag{15}$$

With $C_L = 1\,\text{pF}$ and $\epsilon = 10^{-3}$,

$$t_{0.1\%,initial} = 462.96 \times 10^{-12} \ln(1000) = 3.198\,\text{ns}. \tag{16}$$

Conversely, the 10 ns budget gives the guideline

$$R_{\text{ON}} \lesssim \frac{10\,\text{ns}}{1\,\text{pF}\ln(1000)} = 1.448\,\text{k}\Omega. \tag{17}$$

For loads of 0.1 pF and 20 fF, the same model predicts 0.320 ns and 0.064 ns, respectively. Values this small can no longer be separated from the 1 ns control edge and the internal switch capacitance; the prediction only shows the ideal RC scale.

3.3 Why a single RC constant can miss

The effective source and drain of the TG change with the signal. When both terminals are near mid-supply, the NMOS V_{GS} and PMOS V_{SG} both shrink, while bulk bias raises the magnitude of the effective threshold. The simple body-effect form for an NMOS is

$$V_T = V_{T0} + \gamma \left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right). \tag{18}$$

No value of γ was extracted in this work. The equation explains the direction of the effect; it does not compute the numerical threshold of the final transistors. The illustrative 0.45 V threshold is not sufficient to represent all bias conditions of the standard model used.

If the current varies strongly with voltage, the transition time is better viewed through

$$t = C_L \int_{V_a}^{V_b} \frac{dV}{I_{path}(V)}, \tag{19}$$

with the direction of integration adjusted for charging or discharging. A weakly conducting region can dominate the integral even when the resistance near the rail is low. Therefore the minimum R_{ON} cannot represent the whole transition.

3.4 Reference energy and the cost of the control load

One bit change swaps both outputs. For two equal loads C_L, the charge moved in each branch is

$$\Delta q = C_L \Delta V_{\mathrm{REF}} = 1\,\mathrm{pF}(0.6\,\mathrm{V}) = 0.6\,\mathrm{pC}. \tag{20}$$

The ideal net reference energy for one reversal of the output pair is

$$E_{ref} = V_{\mathrm{REFP}} C_L \Delta V_{\mathrm{REF}} - V_{\mathrm{REFN}} C_L \Delta V_{\mathrm{REF}} = C_L \Delta V_{\mathrm{REF}}^2 = 0.36\,\mathrm{pJ}. \tag{21}$$

If α denotes the probability that the bit changes per sample,

$$P_{ref,ideal} = \alpha f_s C_L \Delta V_{\mathrm{REF}}^2. \tag{22}$$

The result is $1.44\,\mu$W for $\alpha = 1$ and $0.72\,\mu$W for $\alpha = 0.5$. For the initial rail-to-rail 1.8/0 V example, the corresponding figures are 3.24 pJ and 12.96 μW for $\alpha = 1$. This ninefold reduction comes from the *change of reference span*, not from evidence that efficiency improved ninefold through size optimization.

Each Q or QB drives two DAC NMOS and two DAC PMOS devices. The gate-area proxy driven by one node is

$$A_{gate,Q} = 2(W_n + W_p)L. \tag{23}$$

Its value grows from $1.8\,\mu\mathrm{m}^2$ for 2/4 to $79.2\,\mu\mathrm{m}^2$ for 8/256, a factor of 44. Using only the initial illustration $C_{ox,eff} = 5\,\mathrm{fF}/\mu\mathrm{m}^2$, the capacitance proxy changes from 9 to 396 fF per Q/QB. The 396 fF figure is not an extracted capacitance; overlap, channel bias, and routing are not covered.

Likewise, $\sum WL$ of the eight DAC MOS devices increases from 3.6 to $158.4\,\mu\mathrm{m}^2$. This is a channel metric, not layout area. It is useful for comparing candidates but excludes contacts, diffusion, wells, spacing, and metal.

3.5 Glitch and charge on the released node

A simple coupling estimate gives

$$\Delta V_{ft} \sim \frac{C_{couple}}{C_L}\Delta V_{control}, \qquad \Delta V_q \sim \frac{\Delta q_{inj}}{C_L}. \tag{24}$$

The initial example $C_{couple} = 2\,\text{fF}$ gives 3.6 mV at 1 pF and an 1.8 V edge. This is only an arithmetic example. The change in final dimensions and the sum of several mechanisms mean it is not a prediction of the final amplitude. Parasitic gate coupling and channel charge transfer can disturb analog nodes; adding width to reduce resistance also brings a parasitic cost [4].

4 Experimental method and measurement definitions

4.1 From the automation netlist to the Xschem export

Python automation assembles the decks, runs ngspice, reads the waveforms, and tests the criteria. Python does not replace the electrical circuit solution. The electrical results in this report come from ngspice in the user's environment with the SKY130 model, not from a software test fixture. Code inspection and dry runs only validate the assembly and data analysis.

The early stage used a parametric netlist backend. After the size and interface were chosen, the schematic was built in Xschem and exported. The final tests use the fixed sizes from that export. Stage names and result rows distinguish these two phases so that backend results are not simply equated with schematic results.

Table 3: Test sequence and the design question of each.

Stage	Test	Question
4A	Logic and output values	Is the reference selected with the correct polarity?
4B	DC R_{ON}, load, settling	How strong is the switch over the tested signal range?
4C	Edges, skew, node release	What are the transient cost and the control asymmetry?
4D	45 standalone PVT points	Does the nominal size withstand the defined variation?
4E	Comparator–buffer–latch–DAC	Does the complete interface work with transistor drivers?

4.2 DC resistance: small differential measurement

The fixture applies a voltage difference $\delta = 1\,\text{mV}$ symmetrically about the common-mode voltage, then compares the currents for positive and negative polarity:

$$G_{ON}(V_{CM}) = -\frac{I_+(V_{CM}) - I_-(V_{CM})}{2\delta}, \qquad R_{\text{ON}} = G_{ON}^{-1}. \tag{25}$$

The minus sign follows the SPICE source-current convention in the deck. A 61-point sweep covers almost the entire 0–VDD range. Comparing the two polarities removes the first-order common-mode leakage component. The result is still a local measurement at small V_{DS}; it is not a single constant resistance for all transitions.

4.3 Definitions of settle and hold

For each bit, the per-branch target is set by Equation (4). The test error is

$$e(t) = \max\left(|V_P - V_{P,target}|, |V_N - V_{N,target}|\right). \tag{26}$$

The tolerance $e_{tol} = 0.001\Delta V_{\mathrm{REF}}$ is 0.60 mV nominally and 0.54 mV at VDD 1.62 V. Settling is determined from the last sample that still fails, then the next sample that meets the tolerance through the end of the window with a minimum persistence time of 20 ns. The first crossing alone is not sufficient.

With ideal control, the settling start is the completion of both control ramps. With transistor integration, the start is when Q and QB both enter the 1% rail band. The latch decision delay uses a 30%/70% logic threshold, while the comparator is judged by the polarity of its regenerative output. These three metrics are deliberately not merged.

With t_0 the start of the clock edge and t_g the time at which control reaches the rail,

$$t_{DAC,after} = t_{DAC,stable} - t_g, \qquad t_{clock\to DAC} = t_{DAC,stable} - t_0. \tag{27}$$

Adding two maxima from different cases is not a total measurement of one waveform. The total must be computed on the same transition.

4.4 Numerical validity and the limits of PASS status

The final deck uses the trapezoidal method, `reltol=1e-5`, `vntol=1e-8`, `abstol=1e-13`, and a maximum time step of 0.05 ns. The log confirms the 10^{-6} geometry scale and the ngspice-46 simulator. Waveforms are saved with `wrdata`; the transient facilities and control language follow the ngspice documentation [5].

A case that times out or fails to load a Python module is not an electrical result that fails the specification; it must be fixed and rerun until valid data are available. Conversely, an ngspice process that finishes is not evidence that the circuit passes. The pipeline requires a valid execution status, complete data, and function and timing checks. The cache checks the identity of the deck, simulator, and model dependencies so that old results are not used for a changed configuration.

5 Design evolution: results that changed decisions

5.1 Initial 2/4 size: the model was too optimistic

The nominal DC result shows $R_{\mathrm{ON}}(0.9\,\mathrm{V}) = 4098.08\,\Omega$, a minimum of $463.21\,\Omega$, and a maximum of $5802.37\,\Omega$ [S02]. The minimum value, which happens to be close to the initial estimate, does not validate the mid-signal estimate. Comparison at the same mid-point gives

$$\frac{R_{\mathrm{ON}}, sim(0.9\,\mathrm{V})}{R_{\mathrm{ON}}, initial} = \frac{4098.08}{462.96} = 8.85. \tag{28}$$

Settling at the 1.2/0.6 V reference and a 1 pF load reached 17.77 ns, whereas the initial estimate was only 3.20 ns. This difference supports rejecting the illustrative parameters as a quantitative sizing tool. The available data do not separate the contributions of threshold, mobility, and body effect numerically; the physical explanation is consistent with TG bias but is not an extraction of the mechanism.

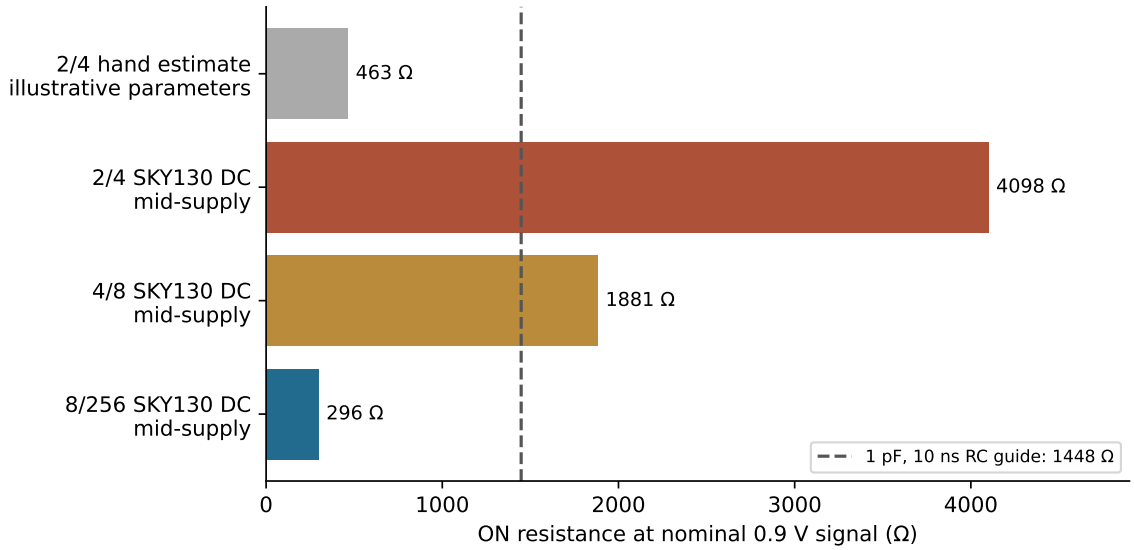


Figure 3: Initial estimate versus DC resistance at the same signal points. The dashed line is the 1 pF/10 ns RC guideline, not a universal threshold for every transition.

5.2 4/8 size: a nominal improvement is not enough

Doubling both widths reduces the mid-point R_{ON} from 4098 to 1881 Ω. The maximum settling of the nominal sweep becomes 8.763 ns; the 4/8 choice has the smallest $\sum WL$ among the candidates meeting that stage [S03]. Test 4A passes 6/6 and 4C completes 54/54: 45 switching transients plus nine release characterizations [S04,S05].

However, PVT passes only 26/45. At SF, 1.62 V, −40 °C, settling grows to 137.923 ns. A reduced time step gives 137.893 ns, a change of about 0.022%, so this large failure does not disappear through that refinement alone [S06,S07]. This result refutes the hypothesis that a nominal pass can be transferred directly to the whole PVT grid.

DC measurement at the failing condition finds a maximum R_{ON} of about 145.18 kΩ. At the same corner, settling for 20 fF, 0.1 pF, and 1 pF is 4.825, 15.662, and 137.923 ns [S08]. The load dependence supports the explanation of limited conduction. The local maximum resistance must not be directly multiplied into a full settling prediction because it does not hold along the entire voltage path.

5.3 Final 8/256 size and the cost of the solution

Worst-condition screening shows that adding NMOS alone is not sufficient: 8/8 still requires 128.70 ns, whereas 8/16 and 8/32 give 70.61 and 38.50 ns [S09]. After the search was extended, the selected 8/256 candidate passes 45/45 with a standalone maximum of 7.084 ns [S10]. This candidate is not evidence of a global optimum. The size grid is limited, and the area objective does not yet include layout cost, power, and all transient limits.

As a retrospective check, assuming $t_s \propto 1/W_p$ from the 8/32 point would give $W_{p,trial} \approx 32(38.50/10) = 123.2\,\mu$m for a 10 ns target. However, the same assumption predicts only $38.50(32/256) = 4.81$ ns at 8/256, faster than the actual result of 7.084 ns. Inverse width scaling therefore helps set the search direction but is not sufficient to determine the final size. Capacitance growing with width and a conduction path that is not purely PMOS may limit the improvement; their contributions were not extracted separately. This retrospective calculation is not claimed to have been made before the search took place.

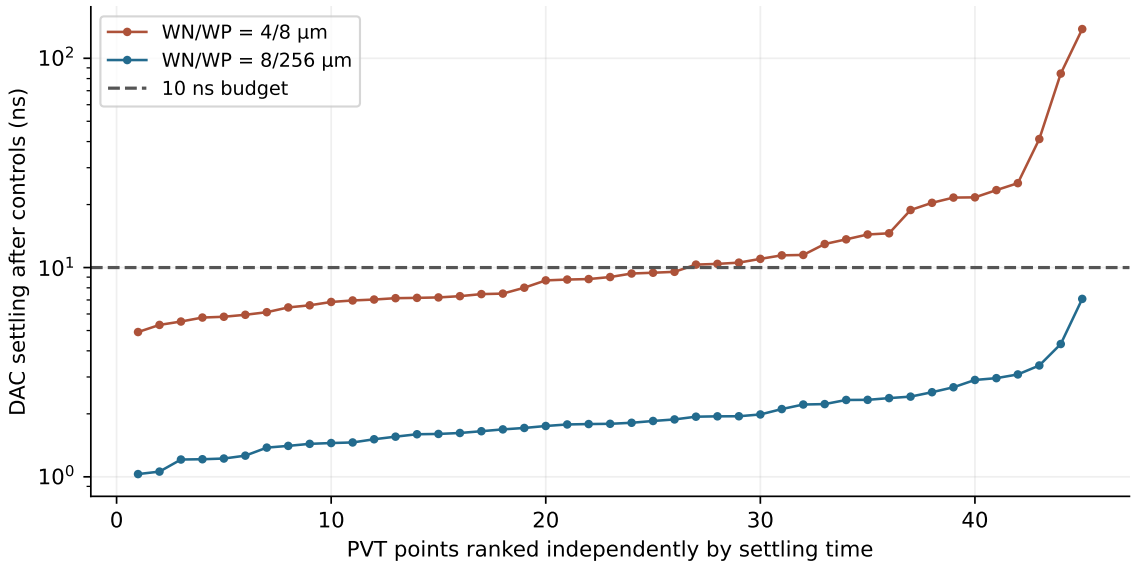


Figure 4: Distribution of settling over the 45 standalone PVT points. The points of each curve are sorted independently; the horizontal axis is not the same corner pair. The 4/8 size fails 19 points, whereas 8/256 meets this entire grid.

At nominal, the final candidate has a mid-point R_{ON} of 296.34 Ω, a maximum of 365.85 Ω, and a maximum 4B sweep settling of 1.880 ns [S11]. The RC model using the maximum resistance gives $365.85\,\Omega \times 1\,\text{pF} \times \ln(1000) = 2.527\,\text{ns}$. The nanosecond scale agrees, but the difference between 2.527 and 1.880 ns remains plausible because the voltage has already moved during the control ramp and the resistance is not constant. The simple model succeeds as a scale check, not as a precise predictor.

5.4 The 1 mV integration problem and loading diagnosis

After the standalone DAC passed, integration without a buffer completed 24 simulations, but only 18 met the criteria. The six failures were at the 1 mV input difference, including the configuration without a DAC [S13]. Blaming the DAC size alone therefore does not explain all of the data.

Table 4: Experiments separating the influence of loading [S14].

Configuration	Comparator correct	Output correct	Hold correct
Original connection, 1 mV	6/12	6/12	6/12
Original connection, half time step	6/12	6/12	6/12
Comparator without latch	12/12	12/12	–
Latch with diagnostic ideal isolation	12/12	12/12	12/12
Original connection, 5 mV	12/12	12/12	12/12

Ideal isolation was used to test the cause, not kept as a physical solution. Together, these results support an influence of the comparator–latch interaction at small signals. The terms loading, back-action, and regeneration describe possible mechanisms; these data do not measure the contribution of each separately.

Four physical buffer sizes B01–B04 were tested and all passed the six 1 mV screening cases. B01 was chosen because its additional channel proxy is the smallest, $1.35\,\mu\text{m}^2$ for both branches. Each buffer uses an inverter with $W_n/W_p = 0.5/1\,\mu\text{m}$ followed by $1/2\,\mu\text{m}$, all with $L = 0.15\,\mu\text{m}$. B01

then passed nominal verification 24/24 [S15]. This selection proves the sufficiency of B01 within the test space, not that a larger buffer would never be needed at other loads.

5.5 Moving the design to an inspectable schematic

The Xschem DAC export passes 4A 6/6 and standalone PVT 45/45, with a maximum of 7.073 ns [S16,S17]. The value is close to the backend 7.084 ns but is reported separately. The complete integration export then passes 24/24 nominal [S18].

The final netlist shows MOS bulks connected, consistent interface pins, two physical B01 buffers, and fixed 8/256 sizes. In the automation deck, the two buffers are wrapped inside a subcircuit named SR_Latch, while the two-NAND core is kept as SR_Latch_CORE. This hierarchical placement does not mean the buffers disappeared, and it does not require adding a second buffer.

6 Final results from the Xschem export

6.1 Campaign coverage and passes

The final campaign contains five corners TT/SS/FF/SF/FS, three supplies 1.62/1.8/1.9 V, and three temperatures −40/27/125 °C. Thus

$$N_{PVT} = 5 \times 3 \times 3 = 45. \tag{29}$$

At each point there are four input-difference magnitudes 1/5/10/50 mV, three patterns alternating/repeats/PRBS, and two configurations loaded/unloaded:

$$N_{case} = 45 \times 4 \times 3 \times 2 = 1080. \tag{30}$$

Of these, 540 cases connect the DAC with a 1 pF load per branch; the other 540 test the comparator–buffer–latch without a DAC. All use a fixed input common mode of 0.9 V; this is not a common-mode sweep. The analog and reference sources are ideal low-impedance sources; the PRBS pattern uses a fixed seed, not random mismatch samples.

All cases have OK execution status and pass the function, settling, and preliminary criteria. There is no unresolved decision, unsettled gate, or unsettled edge. The case count and the uniqueness of point/ID pairs were rechecked from the CSV [S19,S20,S21]. The 1080/1080 figure is deterministic coverage, not an estimate of production yield.

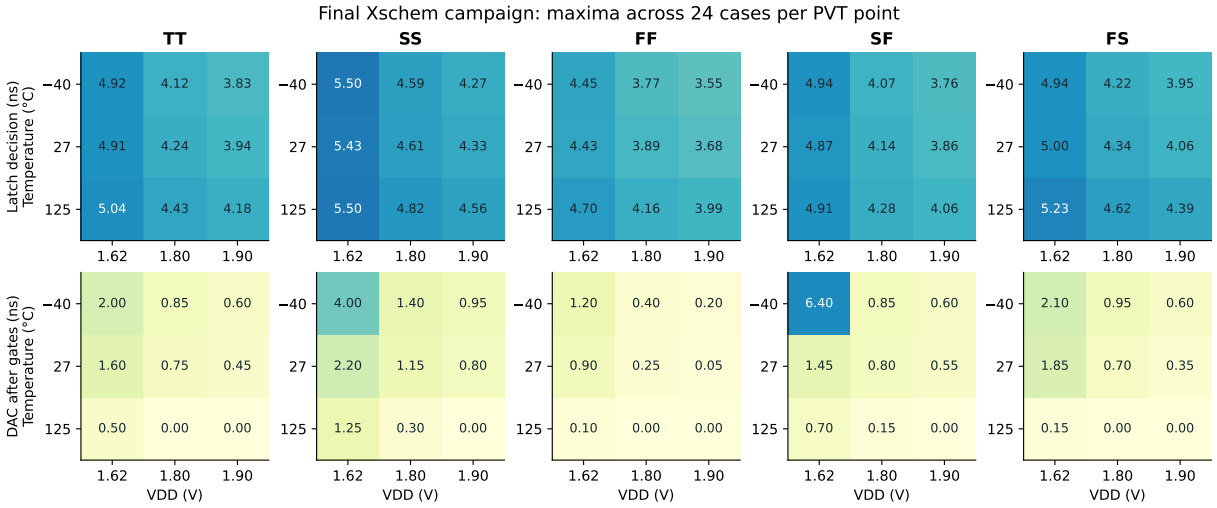


Figure 5: Maximum over the 24 cases at each PVT point. Top row: latch decision delay. Bottom row: DAC settling after control is stable, maximum over the loaded cases. All figures in ns.

6.2 Worst metrics and where they occur

Table 5: Extremes of the final campaign. The values come from different cases and are not summed into a single waveform.

Metric	Maximum	Corner / VDD / temp. / ID
Latch delay	5.498 ns	SS / 1.62 / 125 / 0001
Gate reaching 1% rail	7.853 ns	SS / 1.62 / 125 / 0001
DAC settling after gate	6.400 ns	SF / 1.62 / −40 / 0013
Steady-state error	1.254 μV	SS / 1.62 / 125 / 0015
Excursion beyond reference	38.775 mV	FS / 1.90 / −40 / 0019
Peak reference current	2.207 mA	FF / 1.90 / 27 / 0001
DAC terminal-pair proxy	1.913 V	SF / 1.90 / −40 / 0007
Transient common-mode excursion	229.105 mV	SS / 1.62 / −40 / 0001

A settling time of 6.400 ns leaves 3.600 ns against the 10 ns post-gate budget. The 1% rail limit uses 16.2 mV at VDD 1.62 V. These two kinds of tolerance must not be confused with the 0.54 mV analog DAC tolerance.

6.3 Waveform review: slowest decision

The case SS/1.62 V/125°C, ID 4E_0001, uses an input difference of ±1 mV. The waveform has 62,794 samples up to 3.1 μs. Recomputation gives a latch delay of 5.498 ns, a gate time of 7.853 ns, and an additional DAC settling of 1.200 ns. The maximum total clock-to-DAC time in this waveform is 9.053 ns [S23,S24].

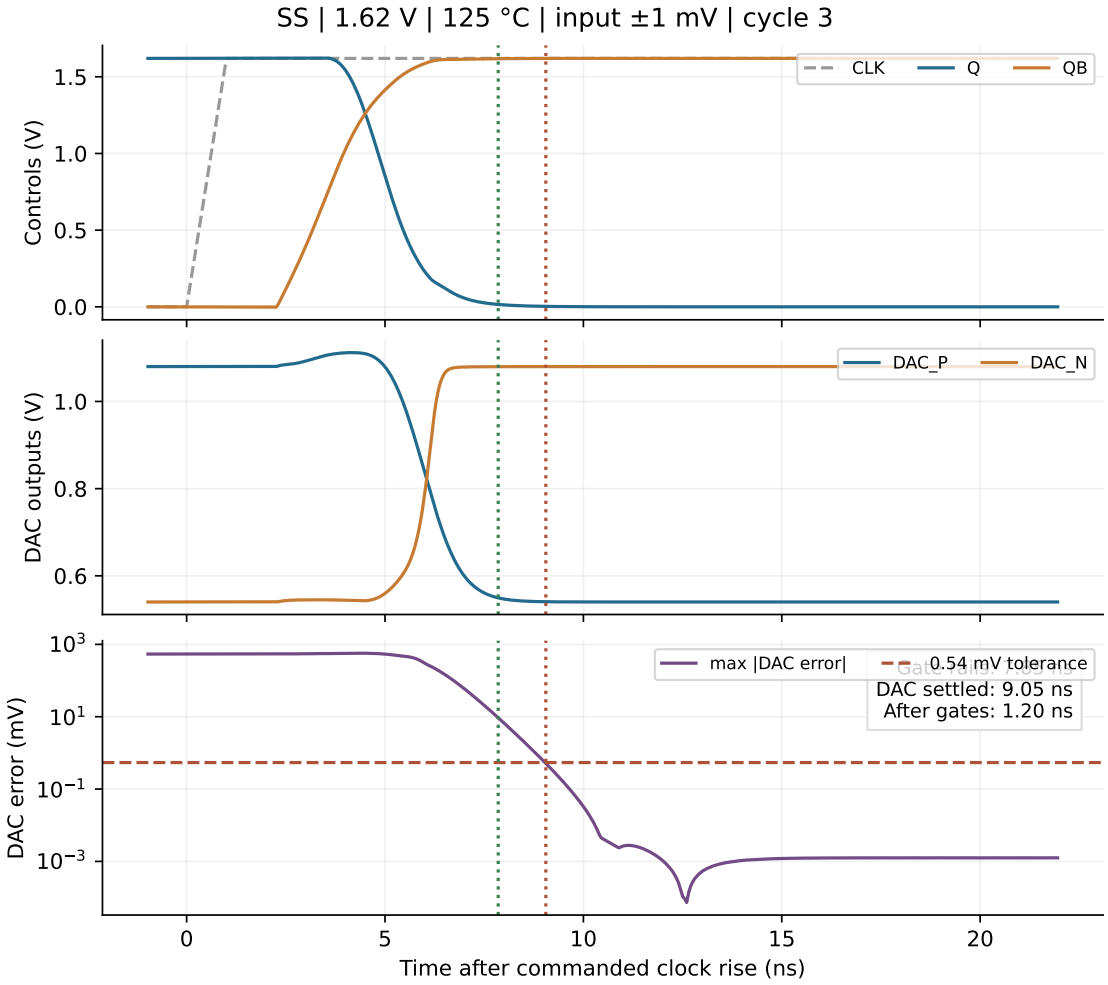


Figure 6: SS case at the third-cycle transition. The green line marks control entering the rail band; the red line shows the DAC entering the tolerance and staying. Time is counted from the start of the clock ramp, not the 50% crossing.

6.4 Waveform review: slowest DAC settling

The case SF/1.62 V/−40°C, ID 4E_0013, uses an input difference of ±10 mV and has 62,963 samples. In the second cycle, the gate reaches the rail at 6.311 ns; the DAC requires a further 6.400 ns. Hence

$$t_{clock \to DAC} = 6.311 + 6.400 = 12.711\,\text{ns}. \tag{31}$$

This figure is the maximum measured over the two reviewed waveforms, not a total maximum computed from all 1080 waveforms. Passing the post-gate budget remains valid, but taking 6.400 ns as the total integration delay would understate the time required.

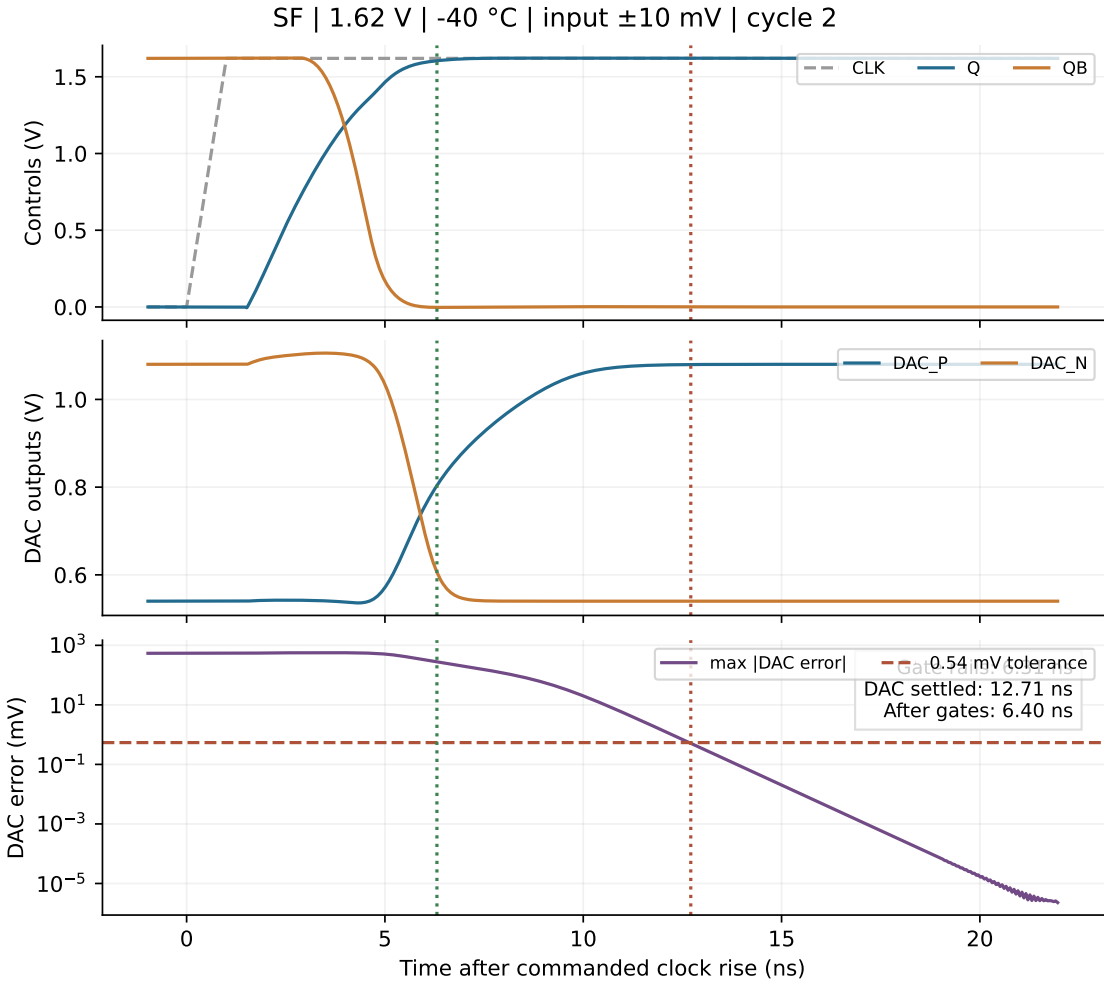


Figure 7: SF case, second cycle. The error curve shows a settling tail that persists after Q/QB have nearly reached the rail.

6.5 Hold during reset and limits of the initial condition

The review extends the reset check from the start of the falling edge to just before the next evaluation. Both waveforms give 12/12 correct decisions and 12/12 reset intervals without a bit change. The maximum disturbance of Q/QB from the rail is 6.169 mV for SS and 4.489 mV for SF, both within the 1% VDD band. The maximum DAC error during reset is 115.769 and 95.281 μV; both are below 540 μV.

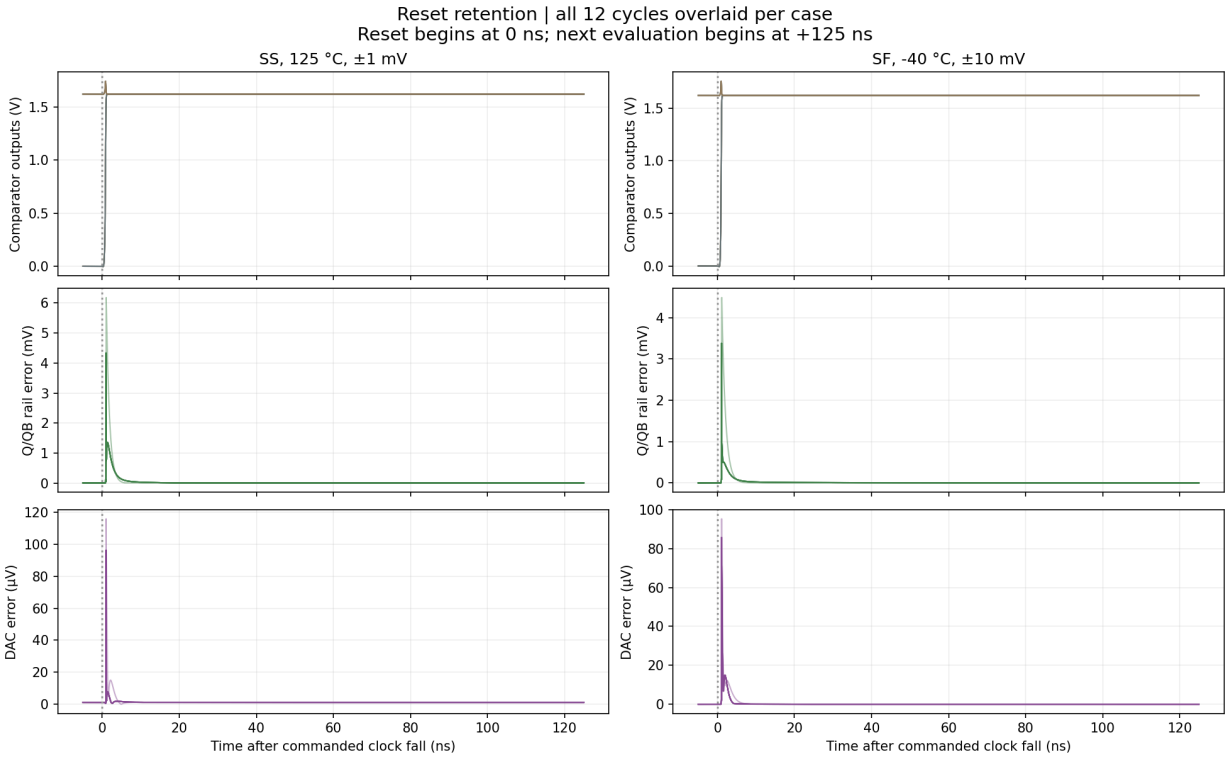


Figure 8: Twelve reset intervals overlaid per case. The comparator returns high; the latch rail error and DAC disturbance decay without a change of decision. The right and left axes have different scales.

The first cycle already has Q=0 from the DC operating-point solution. A zero delay in the first row is therefore not evidence that the circuit has a physical zero delay or a guaranteed random startup. The two waveforms contain 22 bit changes that were actually tested after the two initial states. No warnings, errors, or early terminations were found in either log.

7 Discussion: what agrees and what does not yet

7.1 Reconciling hypotheses with evidence

Table 6: Expectations, results, and updated design understanding.

Expectation	Test result	Explanation / decision
Four TGs give $\pm\Delta V_{\mathrm{REF}}$ outputs with fixed CM in the steady state.	Polarity and steady state agree.	Transfer structure kept; the CM transient must still be assessed.
2/4 suffices because the initial RC is 3.20 ns.	Settling 17.77 ns; mid-point DC 4.10 kΩ.	Illustrative parameters too optimistic; sizing uses the PDK model.
4/8, which passes nominal, suffices for PVT.	Only 26/45; worst 137.92 ns.	Nominal does not represent the corners; worst-case optimization needed.
A standalone circuit that passes can be combined directly.	1 mV fails at the comparator–latch connection.	The dynamic interface must be tested; two B01 give a verified physical solution.
A fixed-size schematic reproduces the netlist solution.	24/24 nominal and 1080/1080 PVT pass.	Xschem export baseline accepted within the test coverage.

The early failure does not mean the RC equation is useless. It gives the correct cause-and-effect relation: a larger load or weaker conduction lengthens settling. What failed was the use of a single resistance value that did not yet represent the actual transistors and bias.

7.2 Power: aligning the domains before comparing

The engine power metric integrates the net energy from VDD and both references; with ideal control drivers, the Q/QB source power is also counted. In integration, the power includes the comparator, buffers, latch, and DAC. It is not DAC power alone.

As a measured example, TT/1.8 V/27°C, 1 mV alternating loaded, has an average source power of 7.661 μW. The earlier ideal reference-load estimate was only 1.44 μW at full activity. The difference must not be called a fivefold theory error: the source coverage differs, there is gate and decision-circuit activity, and the 100–3100 ns window contains a limited bit sequence. The maximum over the entire campaign is 12.033 μW, also for the full interface [S19].

7.3 Non-idealities that cannot yet be labeled as passing

The output excursion beyond the reference range reaches 38.775 mV in the final campaign. This metric does not count the whole difference from the target while the intended transition is still in progress. The peak reference current of 2.207 mA includes load charging, parasitics, and possible overlap; it is not a pure shoot-through measurement.

Table 7: Transient cost in the standalone 4C screening, size 8/256 [S12].

Load per output	Switching cases	Max. excursion (mV)	Max. settling (ns)
20 fF	15	221.973	0.444
0.1 pF	15	185.545	0.558
1 pF	15	65.493	1.913

This screening combines three control edges (0.2, 1, 5 ns) and five skews (−2, −0.5, 0, 0.5, 2 ns). An ideal Q/QB source can sustain a different overlap than a transistor driver. For that reason, the 221.973 mV maximum at light load and the maximum current of about 8.93 mA at that stage must not be combined with the final campaign maxima as a single operating condition. The result is still important: a light load settles faster but does not automatically see a smaller glitch.

The backend release test of the final size at 1 pF gives a change of 38.65–41.04 mV, or a $C_L\Delta V$ proxy of about 38.65–41.04 fC [S12]. The node is deliberately released from the reference so that the disturbance is stored. This differs from a DAC that remains connected to the reference after switching. The proxy mixes injection, feedthrough, and drift; it does not extract pure channel charge. A characterization status that passes while the glitch limit is empty does not prove that the system charge-error limit is met.

A correct steady-state common mode also does not erase a transient excursion of up to 229.105 mV. In subsequent circuits its effect depends on the sampling phase, the impedance, and the OTA response. Integration with the actual SC load must therefore check when this transient becomes visible to the sensitive capacitors.

7.4 Model validity is not a reliability sign-off

The SKY130 documentation gives the model bias range for 1.8 V devices, including magnitude ranges of V_{GS} and V_{DS} up to 1.95 V [2]. That range does not automatically become a lifetime limit applicable to every terminal pair. The 1.913 V maximum proxy covers only DAC MOS terminal pairs. Signed bias against the bulk and the internal comparator/buffer/latch transistors are not yet covered in the full stress decision.

Three final configuration limits are still `null`: maximum glitch, maximum peak reference current, and terminal-voltage limit. Because they are not set, a `preliminary_pass=True` result must not be written as passing all specifications. Likewise, the 0.1% settling tolerance corresponds to an amplitude ratio of 10^{-3}; it is not evidence of system dynamic range (initial target 99 dB, revised to about 75 dB at OSR 80). The system error and noise allocation has not been derived from that target.

8 Conclusion and the baseline retained

Block 4 has reached a pre-layout baseline that can be traced from theory, calculation, failure, and diagnosis to schematic export. The four TGs still implement the desired transfer. The main changes are the enlargement of the devices to 8/256 and the addition of two B01 buffers at the comparator–latch interface.

The strength of this result lies in the order of proof: illustrative parameters were tested and rejected when they did not agree; a nominal pass was not used to hide a PVT failure; ideal sources were used only for diagnosis; and the final solution was verified again with transistors and the Xschem export. All 1080 cases meet the defined campaign criteria.

This size is not called a global optimum and has not been translated into a layout that passes DRC/LVS/PEX. The next work uses the same baseline to assess the interaction with the switch network and capacitors of Block 5. New checks should answer a change of load or interface assumption, not repeat the entire sweep without reason.

Statement of result usable in a portfolio. A differential one-bit SKY130 feedback DAC was designed and verified at the transistor level, including diagnosis of nominal/PVT failures, repair of the comparator–latch interface, and retesting of the Xschem export at 45 PVT points and 1080 deterministic cases. This claim is limited to pre-layout function and timing according to the documented criteria.

A Reproducibility and final configuration

The model recorded in the deck uses the SKY130A installation through Ciel with the revision identity

```
7b70722e33c03fcb5dabcf4d479fb0822d9251c9
```

This identity refers to the installation that was tested, not a claim that all versions of the online documentation are exactly the same. The PDK model is not redistributed in the report package.

Final campaign command in the user's automation directory:

```
python3 stage_4E_buffer_pvt.py \
  --config 4E_Xschem_Verification/config_nominal.json \
  --dac-netlist 4E_Xschem_Verification/dac_xschem.inc \
  --block3-netlist 4E_Xschem_Verification/block3_xschem_b01.inc \
  --all-inputs \
  --pdk /foss/pdks/sky130A/libs.tech/combined/sky130.lib.spice \
  --workers 2 --out results/4E_xschem_pvt_full
```

This package includes snapshots of the engine and runner to explain the method, not the PDK or the complete simulation environment. Netlist/config files are in `sources/`; result data are in `data/`. Original names and SHA-256 are recorded in `data/source_manifest.json`. A ZIP of the two raw waveforms, the pre-simulation note, the per-cycle tables, and the review results are also included as evidence that can be rechecked.

To compile the report, run XeLaTeX twice, for example `xelatex main_en.tex`, or upload the package to Overleaf and select the XeLaTeX compiler. The Indonesian version uses `main_id.tex`. Both documents use the same numerical evidence.

B Index of experimental evidence

Evidence	Content and role
S00	Design note written before simulation
S01–S02	2/4 baseline: logic, resistance, and settling
S03–S05	Nominal search and verification of the 4/8 size
S06–S09	PVT failure, refinement, and conduction diagnosis
S10–S12	8/256 size: PVT, resistance, skew, release
S13–S14	Unbuffered integration and isolation experiments
S15	B01–B04 screening and nominal verification of B01
S16–S17	Logic and standalone PVT from the Xschem DAC export
S18	Nominal integration from the Xschem export, 24 cases
S19	All 1080 result rows of the final integration PVT campaign
S20–S21	Aggregation over 45 points and final campaign summary
S22	User-owned integration netlist export
S23	Two worst cases: waveforms, decks, configuration, logs
S24	Independent waveform review and per-cycle tables

C Summary of the final 45 PVT points

The values below are maxima per point, not a single transition containing all maxima at once. Each point contains 24 cases and all pass the campaign criteria.

Corner	VDD (V)	Temp. (°C)	Decision (ns)	DAC (ns)
TT	1.62	-40	4.924	2.000
TT	1.62	27	4.911	1.600
TT	1.62	125	5.041	0.500
TT	1.80	-40	4.122	0.850
TT	1.80	27	4.236	0.750
TT	1.80	125	4.435	0.000
TT	1.90	-40	3.834	0.600
TT	1.90	27	3.938	0.450
TT	1.90	125	4.176	0.000
SS	1.62	-40	5.497	4.000
SS	1.62	27	5.428	2.200
SS	1.62	125	5.498	1.250
SS	1.80	-40	4.585	1.400
SS	1.80	27	4.606	1.150
SS	1.80	125	4.822	0.300
SS	1.90	-40	4.269	0.950
SS	1.90	27	4.332	0.800
SS	1.90	125	4.556	0.000
FF	1.62	-40	4.450	1.200
FF	1.62	27	4.430	0.900
FF	1.62	125	4.696	0.100
FF	1.80	-40	3.770	0.400
FF	1.80	27	3.889	0.250
FF	1.80	125	4.162	0.000
FF	1.90	-40	3.551	0.200
FF	1.90	27	3.678	0.050
FF	1.90	125	3.992	0.000
SF	1.62	-40	4.945	6.400
SF	1.62	27	4.874	1.450
SF	1.62	125	4.908	0.700
SF	1.80	-40	4.074	0.850
SF	1.80	27	4.142	0.800
SF	1.80	125	4.276	0.150
SF	1.90	-40	3.764	0.600
SF	1.90	27	3.858	0.550
SF	1.90	125	4.060	0.000
FS	1.62	-40	4.943	2.100
FS	1.62	27	4.999	1.850
FS	1.62	125	5.227	0.150
FS	1.80	-40	4.218	0.950
FS	1.80	27	4.342	0.700
FS	1.80	125	4.624	0.000
FS	1.90	-40	3.946	0.600
FS	1.90	27	4.058	0.350
FS	1.90	125	4.388	0.000

References

[1] B. E. Boser and B. A. Wooley, "The Design of Sigma-Delta Modulation Analog-to-Digital Converters," *IEEE Journal of Solid-State Circuits*, vol. 23, no. 6, pp. 1298–1308, 1988. DOI: 10.1109/4.90025.

[2] SkyWater PDK documentation, "Device Details: 1.8 V NMOS FET and 1.8 V PMOS FET." https://skywater-pdk.readthedocs.io/en/main/rules/device-details.html. Accessed 15 October 2026. Online documentation is contextual; the tested model revision is identified in Appendix A.

[3] Analog Devices, "Selecting the Right CMOS Analog Switch." https://www.analog.com/en/resources/design-notes/selecting-the-right-cmos-analog-switch.html. Accessed 15 October 2026.

[4] M. McCarthy and A. Collins, "Ask The Applications Engineer—26: Switches and Multiplexers," *Analog Dialogue*, Analog Devices. https://www.analog.com/en/resources/analog-dialogue/articles/ask-the-applications-engineer-26.html. Accessed 15 October 2026.

[5] Ngspice project, *Ngspice User's Manual* and control-language documentation. https://ngspice.sourceforge.io/docs.html; https://ngspice.sourceforge.io/ngspice-control-language-tutorial.html. Accessed 15 October 2026. Experiment logs identify simulator version 46.